

Problem H. Points on a Rectangle

OS Linux

You are given q queries where each query consists of a set of n points on a two-dimensional plane (i.e., (x, y)). For each set of points, print YES on a new line if all the points fall on the edges (i.e., sides and/or corners) of a [non-degenerate rectangle](#) which is axis parallel; otherwise, print NO instead.

Input Format

The first line contains a single positive integer, q , denoting the number of queries. The subsequent lines describe each query in the following format:

- 1. The first line contains a single positive integer, n , denoting the number of points in the query.
- 2. Each line i of the n subsequent lines contains two space-separated integers describing the respective values of x_i and y_i for the point at coordinate (x_i, y_i) .

Constraints

- $1 \leq q \leq 10$
- $1 \leq n \leq 10$
- $-10^4 \leq x, y \leq 10^4$

Output Format

For each query, print YES on a new line if all n points lie on the edges of some non-degenerate rectangle which is axis parallel; otherwise, print NO instead.

Input	Output
2	YES NO
3	
0 0	
0 1	
1 0	
4	
0 0	
0 2	
2 0	
1 1	

Explanation

We perform the following $q = 2$ queries:

1. In the first query, all points lie on the edges of a non-degenerate rectangle with corners at $(0, 0)$, $(0, 1)$, $(1, 0)$, and $(1, 1)$. Thus, we print **YES** on a new line.
2. In the second query, points $(0, 0)$, $(0, 2)$, and $(2, 0)$ *could* all lie along the edge of some non-degenerate rectangle, but point $(1, 1)$ would have to fall *inside* that rectangle. Thus, we print **NO** on a new line.